

Power Series and Interval of Convergence



Finding the radius of convergence

- 1 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n}$ using the Ratio Test.
 - 2 Find the radius of convergence of $\sum_{n=0}^{\infty} \frac{x^n}{n!}$.
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Finding the interval of convergence (testing endpoints)

- 3 Find the full interval of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n}$ (radius $R = 1$ found earlier), by testing $x = 1$ and $x = -1$.
 - 4 Find the interval of convergence of $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n^2}$.
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Representing functions as power series

- 5 Write the power series for $\frac{1}{1-x}$ using the geometric series formula, and state its interval of convergence.
 - 6 Use the geometric series for $\frac{1}{1-x}$ to find a power series for $\frac{1}{1+x^2}$.
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Differentiating and integrating power series

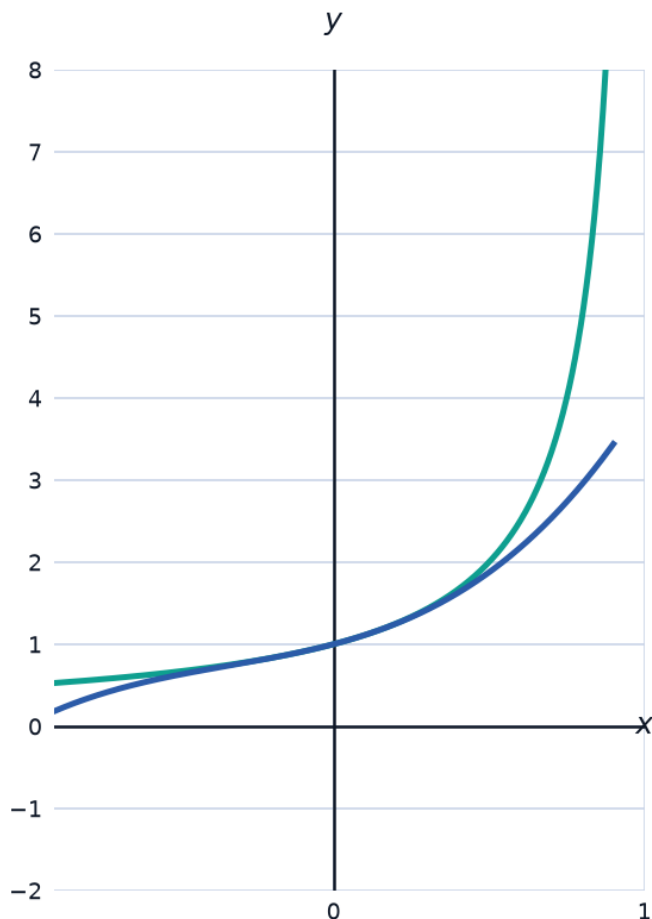
- 7 Using $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$, find a power series for $\frac{1}{(1-x)^2}$ by differentiating term by term.
 - 8 Using $\frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-1)^n x^{2n}$, find a power series for $\tan^{-1} x$ by integrating term by term.
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More radius/interval of convergence practice

- 9 Find the radius and interval of convergence of $\sum_{n=0}^{\infty} n! x^n$.
 - 10 Find the radius and interval of convergence of $\sum_{n=1}^{\infty} \frac{(-1)^n (x+1)^n}{n \cdot 3^n}$.
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Visualizing a power series approximation

- 11 The geometric series $\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots$ converges on $(-1, 1)$. Compare the graph of $f(x) = \frac{1}{1-x}$ to its degree-3 partial sum $P_3(x) = 1 + x + x^2 + x^3$ on $[-0.8, 0.8]$, and describe where they agree most closely.



Term-by-term operations preserve the radius of convergence

- 12 State the theorem describing what happens to the radius of convergence when a power series is differentiated or integrated term by term.
- 13 Given that $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ has radius of convergence $R = 1$, state the radius of convergence of its derivative $\sum_{n=1}^{\infty} x^{n-1}$ without recomputing.

Building a new series by combining known ones

- 14 Using $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ and $\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n$, find the power series for $\frac{2}{1-x^2}$ by adding the two series.

Ratio Test with factorial and power combinations

15 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{2^n x^n}{n^2}$.

16 For the series above, test both endpoints $x = \frac{1}{2}$ and $x = -\frac{1}{2}$ to find the full interval of convergence.

Using a power series to approximate a definite integral

17 Using the power series $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$ found earlier, estimate $\tan^{-1}(0.5)$ using the first three terms.

Multiple choice — easy 15

18 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{6^n}$.

a. 12

b. $\frac{1}{6}$

c. 6

d. 1

19 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{9^n}$.

a. 9

b. $\frac{1}{9}$

c. 18

d. 1

20 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{18^n}$.

a. 1

b. 18

c. 36

d. $\frac{1}{18}$

21 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{13^n}$.

a. $\frac{1}{13}$

b. 26

c. 1

d. 13

22 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{5^n}$.

a. 1

b. 10

c. 5

d. $\frac{1}{5}$

23 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{7^n}$.

a. $\frac{1}{7}$

b. 1

c. 14

d. 7

24 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{20^n}$.

a. 1

b. 20

c. 40

d. $\frac{1}{20}$

25 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{8^n}$.

a. $\frac{1}{8}$

b. 8

c. 16

d. 1

26 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{19^n}$.

a. $\frac{1}{19}$

b. 38

c. 19

d. 1

27 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{14^n}$.

a. 28

b. 14

c. $\frac{1}{14}$

d. 1

28 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{2^n}$.

a. 1

b. 4

c. $\frac{1}{2}$

d. 2

29 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{4^n}$.

a. 4

b. $\frac{1}{4}$

c. 1

d. 8

30 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{12^n}$.

a. 1

b. 24

c. $\frac{1}{12}$

d. 12

31 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{15^n}$.

a. 1

b. $\frac{1}{15}$

c. 15

d. 30

32 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{3^n}$.

a. $\frac{1}{3}$

b. 3

c. 6

d. 1

Multiple choice — medium 15

33 Find the radius of convergence of $\sum_{n=1}^{\infty} 7^n(x+6)^n$.

a. $\frac{1}{7}$

b. 7

c. $\frac{8}{7}$

d. $\frac{1}{8}$

34 Find the radius of convergence of $\sum_{n=1}^{\infty} 7^n(x-4)^n$.

a. $\frac{1}{8}$

b. $\frac{1}{7}$

c. $\frac{8}{7}$

d. 7

35 Find the radius of convergence of $\sum_{n=1}^{\infty} 3^n(x+9)^n$.

a. $\frac{4}{3}$

b. $\frac{1}{4}$

c. $\frac{1}{3}$

d. 3

36 Find the radius of convergence of $\sum_{n=1}^{\infty} 7^n(x+3)^n$.

a. 7

b. $\frac{8}{7}$

c. $\frac{1}{7}$

d. $\frac{1}{8}$

37 Find the radius of convergence of $\sum_{n=1}^{\infty} 5^n(x+1)^n$.

a. $\frac{1}{5}$

b. 5

c. $\frac{6}{5}$

d. $\frac{1}{6}$

38 Find the radius of convergence of $\sum_{n=1}^{\infty} 2^n(x+7)^n$.

a. $\frac{1}{3}$

b. $\frac{3}{2}$

c. $\frac{1}{2}$

d. 2

39 Find the radius of convergence of $\sum_{n=1}^{\infty} 9^n(x-4)^n$.

a. $\frac{1}{10}$

b. $\frac{10}{9}$

c. $\frac{1}{9}$

d. 9

40 Find the radius of convergence of $\sum_{n=1}^{\infty} 2^n(x+4)^n$.

a. $\frac{1}{2}$

b. 2

c. $\frac{3}{2}$

d. $\frac{1}{3}$

41 Find the radius of convergence of $\sum_{n=1}^{\infty} 9^n(x-1)^n$.

a. $\frac{1}{9}$

b. 9

c. $\frac{10}{9}$

d. $\frac{1}{10}$

42 Find the radius of convergence of $\sum_{n=1}^{\infty} 2^n(x+5)^n$.

a. 2

b. $\frac{3}{2}$

c. $\frac{1}{2}$

d. $\frac{1}{3}$

43 Find the radius of convergence of $\sum_{n=1}^{\infty} 2^n(x+8)^n$.

a. $\frac{1}{2}$

b. $\frac{3}{2}$

c. $\frac{1}{3}$

d. 2

44 Find the radius of convergence of $\sum_{n=1}^{\infty} 3^n(x+6)^n$.

a. $\frac{1}{4}$

b. 3

c. $\frac{4}{3}$

d. $\frac{1}{3}$

45 Find the radius of convergence of $\sum_{n=1}^{\infty} 7^n(x+8)^n$.

a. $\frac{8}{7}$

b. $\frac{1}{7}$

c. 7

d. $\frac{1}{8}$

46 Find the radius of convergence of $\sum_{n=1}^{\infty} 8^n(x-5)^n$.

a. $\frac{9}{8}$

b. $\frac{1}{8}$

c. $\frac{1}{9}$

d. 8

47 Find the radius of convergence of $\sum_{n=1}^{\infty} 6^n(x+3)^n$.

a. $\frac{1}{6}$

b. 6

c. $\frac{1}{7}$

d. $\frac{7}{6}$

Multiple choice — hard 15

48 Find the full interval of convergence of $\sum_{n=1}^{\infty} \frac{(x+7)^n}{3^n}$.

a. [-10,-4]

b. [-10,-4)

c. (-10,-4)

d. (-10,-4]

49 Find the full interval of convergence of $\sum_{n=1}^{\infty} \frac{(x-1)^n}{4^n n}$.

a. $[-3,5)$

b. $(-3,5]$

c. $(-3,5)$

d. $[-3,5]$

50 Find the full interval of convergence of $\sum_{n=1}^{\infty} \frac{(x-8)^n}{6^n}$.

a. $(2,14)$

b. $[2,14]$

c. $[2,14)$

d. $(2,14]$

51 Find the full interval of convergence of $\sum_{n=1}^{\infty} \frac{(x-3)^n}{6^n n}$.

a. $(-3,9]$

b. $(-3,9)$

c. $[-3,9)$

d. $[-3,9]$

52 Find the full interval of convergence of $\sum_{n=1}^{\infty} \frac{(x-5)^n}{4^n n}$.

a. $(1,9]$

b. $[1,9)$

c. $[1,9]$

d. $(1,9)$

53 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{n^2 \cdot 8^n x^n}{12^n}$.

a. $\frac{5}{2}$

b. 96

c. $\frac{2}{3}$

d. $\frac{3}{2}$

54 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{n^2 \cdot 8^n x^n}{16^n}$.

a. 2

b. 3

c. $\frac{1}{2}$

d. 128

55 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{n^2 \cdot 5^n x^n}{7^n}$.

a. $\frac{5}{7}$

b. 35

c. $\frac{12}{5}$

d. $\frac{7}{5}$

56 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{n^2 \cdot 6^n x^n}{11^n}$.

a. $\frac{6}{11}$

b. $\frac{17}{6}$

c. $\frac{11}{6}$

d. 66

- 57 Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{n^2 \cdot 9^n x^n}{15^n}$.
- a. $\frac{8}{3}$
 - b. 135
 - c. $\frac{3}{5}$
 - d. $\frac{5}{3}$
- 58 Using $\frac{1}{1-3x} = \sum_{n=0}^{\infty} 3^n x^n$, find the coefficient of x^3 .
- a. 27
 - b. 26
 - c. 81
 - d. 9
- 59 Using $\frac{1}{1-3x} = \sum_{n=0}^{\infty} 3^n x^n$, find the coefficient of x^5 .
- a. 81
 - b. 729
 - c. 15
 - d. 243
- 60 Using $\frac{1}{1-4x} = \sum_{n=0}^{\infty} 4^n x^n$, find the coefficient of x^7 .
- a. 28
 - b. 16384
 - c. 4096
 - d. 65536

61 Using $\frac{1}{1-5x} = \sum_{n=0}^{\infty} 5^n x^n$, find the coefficient of x^3 .

a. 25

b. 125

c. 15

d. 625

62 Using $\frac{1}{1-5x} = \sum_{n=0}^{\infty} 5^n x^n$, find the coefficient of x^7 .

a. 78125

b. 35

c. 390625

d. 15625